

Exp.No: 01

Pg:

Date: 3/2/26

VERIFICATION OF KIRCHHOFF'S LAW

AIM:

To verify Kirchhoff's voltage law and Kirchhoff's current law both theoretically and practically for a given DC circuit.

APPARATUS REQUIRED:

SL.NO	APPARATUS	SPECIFICATION	QUANTITY
1	Regulated Power supply (RPS)	(0-30V)	1
2	Voltmeter	(0-30V) MLC	3
3	Ammeter	(0-10mA) MC	3
4	Resistors	1 K Ω	3
5	Bread Board	—	1

PROCEDURE:-

- 1) Give connections as per the circuit diagram.
- 2) Switch on the supply, vary the RPS to set a particular input voltage.
- 3) Note down the readings of ammeter & voltmeter & tabulate them.
- 4) Vary the RPS for different input voltages and note down the readings of all the meters.
- 5) Reduce the RPS to its minimum value and switch OFF the supply.
- 6) Using the tabulated values, verify Kirchhoff's laws practically, and verify it theoretically.

Mark Splitup

S.No	DESCRIPTION	MARK ALLOTTED	MARK OBTAINED
1	Experimental setup / Connections	20 Marks	20
2	Observation / Readings.	20 marks.	20
3	Calculations	20 marks	20
4	Viva	20 marks	20
5	Record	20 marks.	20
		Total	100

100

10/2/26

Result:-

Thus, Kirchhoff's current law and Kirchhoff's voltage law are verified practically and theoretically.

TABULAR COLUMN :-

V	V ₁	V ₂	V ₃	V = V ₁ + V ₂ + V ₃	I	I ₁	I ₂	I = I ₁ + I ₂
(volts)	(volts)	(volts)	(volts)	(volts)	(amps)	(amps)	(amps)	(amps)
3	1	1.03	1.04	3.0	2.63	1.02	1.03	2.05
6	1.95	2.01	6.05	10.17	4.02	2.01	2.62	4.03
9	2.89	3.08	8.97	14.94	6.03	3.01	3.62	6.3

THEORETICAL CALCULATION

For V = 3V

$$R_{eq} = 1 + 1 + 1 = 3K\Omega, \text{ By Ohms law } V = IR \Rightarrow I = \frac{V}{R_{eq}} = \frac{3}{3K\Omega} = 1mA$$

$$V_1 = R_1 \times I = 1 \times 10^3 \times 1 \times 10^{-3} = 1V, V_2 = R_2 I = 1V, V_3 = R_3 I = 1V$$

$$V = V_1 + V_2 + V_3 = 1V + 1V + 1V = 3V$$

For 6V,

$$R_{eq} = 3K\Omega$$

$$I = \frac{V}{R} = \frac{6}{3} = 2mA$$

$$V_1 = R_1 I = 2V$$

$$V_2 = R_2 I = 2V$$

$$V_3 = R_3 I = 2V$$

For V = 9V

$$R_{eq} = 3K\Omega$$

$$I = \frac{V}{R} = \frac{9}{3K\Omega} = 3mA$$

$$V_1 = R_1 I = 3V$$

$$V_2 = R_2 I = 3V$$

$$V_3 = R_3 I = 3V$$

KCL, For V = 3V

$$I = \frac{V}{R_1} = 2mA, R_{eq} = \frac{1 \times 1}{1+1} = 0.5K\Omega \Rightarrow R_{eq} = 0.5 + 1 = 1.5K\Omega$$

$$I_1 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 2 \times 10^3 = 1.0mA, I_2 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 2 \times 10^3 = 1.0mA$$

For V = 6V

$$I = \frac{V}{R} = 4mA, R_{eq} = \frac{1 \times 1}{1+1} = \frac{1}{2} = 2 \times 10^{-3} \times \frac{1 \times 10^3}{2 \times 10^3} = 1.0mA$$

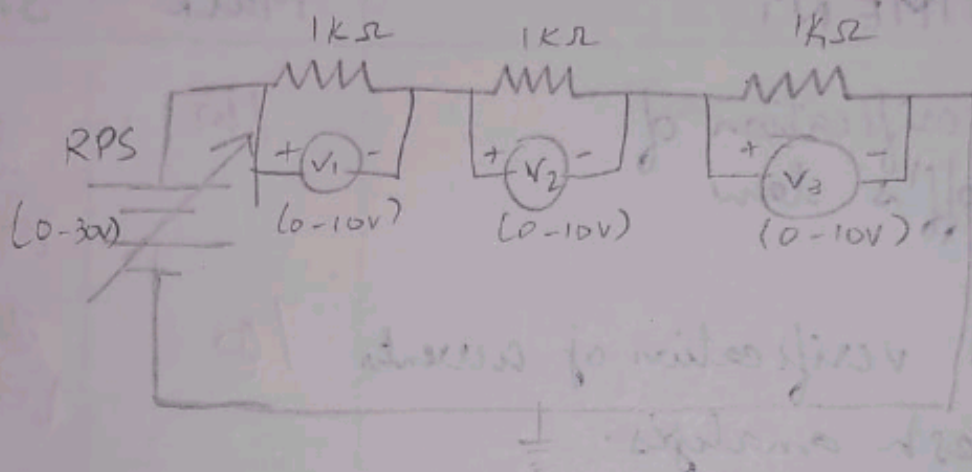
$$I_1 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 4 \times 10^3 = 2mA, I_2 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 4 \times 10^3 = 2mA$$

For V = 9V, $I = \frac{V}{R} = 6mA$

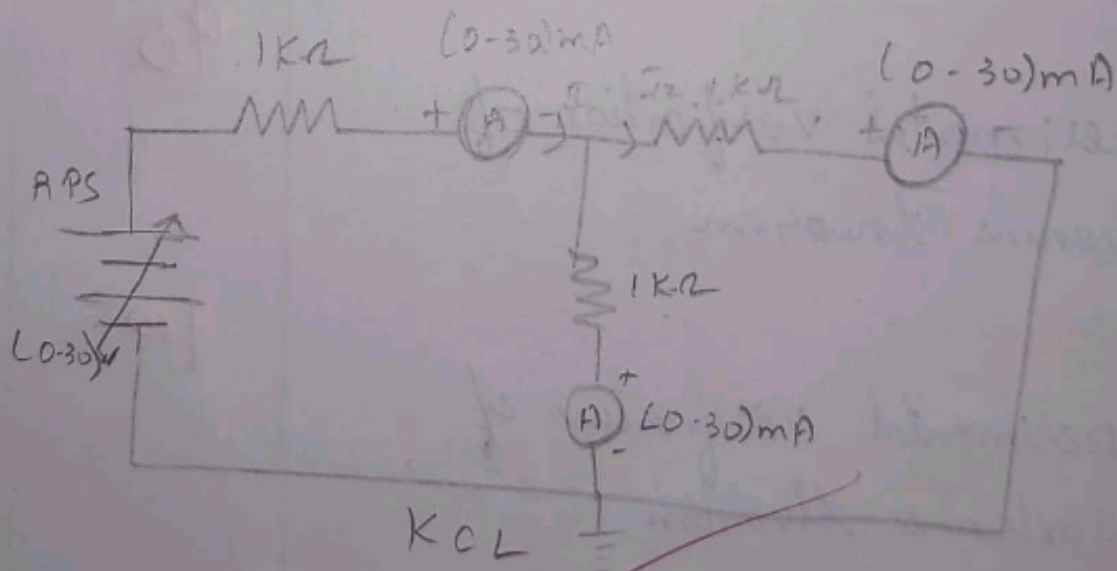
$$I_1 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 6 \times 10^3 = 3.0mA, I_2 = I \times \frac{1K\Omega}{1+1K} = \frac{1 \times 10^{-3}}{2 \times 10^3} \times 6 \times 10^3 = 3.0mA$$

$$R_{eq} = \frac{1 \times 1}{1+1} = 0.5K\Omega \Rightarrow 0.5 + 1 = 1.5K\Omega$$

Circuit Diagram:



KVL



KCL